

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

Register Number	192511426
Name	K.B.Devarshan

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.(BL3)

Assessment Tool Weightage: 10%

QUESTIONS:2

Total Marks:20

Descriptive Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behaviour and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.
2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem.

Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

Code of Conduct:

I _K.B.Devarshan/192511426_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

Criterion	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Understanding of Concepts	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions	5
Explanation	Detailed, well- explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation	5
Application	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis	4
Organization	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers	3
Accuracy	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps	2
Clarity	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand	1

ANSWERS:

Q1.

To design an effective smart home system, we must understand how different agent architectures process real-time inputs

1.SIMPLE REFLEX AGENT:

This agent acts based on the current percept without any memory of the past events. It operates based on condition--action rules.

- **Function in the system:** If the motion sensor detects movement , it turns on the light. If the temperature drops below 68 degree F. ,it turns on the heater.
- **Limitation:** It cannot handle unknown states .If a user sits still reading a book , the motion sensor registers no movement, and the agent turns off the light , causing disturbance.

2.MODEL BASED REFLEX AGENT:

This agent maintains internal state based on percept's history , also tracks the real time environment

- **Function in the system:** If the agent registers that the user has entered the room , keeps the function running until the user exits the room or the operation that has been assigned to the agent .It will not end the task even if the user sits still reading the book, the agent knows the room is occupied & keeps the light on.

3.GOAL-BASED AGENT:

This agent doesn't stop with reacting to the environment .It will get assigned a goal by the user . Whether the agent will achieve the goal or it will make take the process closer to the goal.

- **Function in the system:**

If the user assigns a task "keep the house secure", whether a door or a window is opened without the user's knowledge , here the agent will not only trigger the security alarm , it will ensure the safety by engaging security lock and high tensile motion tracking to ensure the safety of the house and works beyond the user's command.

JUSTIFICATION:

The most effective approach is a Learning Utility-Based Agent.

A standard utility agent handles the complex trade-offs between energy efficiency, comfort, and security. However, adding a learning component is crucial because user preferences change over time. The learning element monitors user corrections (e.g., the user manually turning down the AC when it's too cold) and updates the utility function. This hybrid approach ensures the system doesn't just follow static rules, but actively optimizes for the specific lifestyle of the inhabitants while minimizing energy waste.

Q2.

UNINFORMED SEARCH STRATEGIES

1. Uniform Cost Search (UCS)

UCS expands the node with the lowest path cost from the start node. It explores the maze in a series of expanding possibilities based on accumulated cost.

- If the maze has terrain with different movement costs (e.g., smooth floors cost 1 unit of battery, rough terrain costs 3 units), UCS ensures the robot finds the absolute cheapest path to the exit, not just the path with the fewest steps.
- **Time Complexity:** C^* is the cost of the optimal solution and ϵ is the minimum step cost.
- **Advantages/Disadvantages:** It is complete and optimal. However, it requires a massive amount of memory because it must store all explored nodes in a priority queue.

2. Iterative Deepening Search (IDS)

IDS repeatedly applies Depth-First Search (DFS) with an increasing depth limit (0, then 1, then 2, etc.) until the goal is found.

- It helps to explore path by path like DFS, but by resetting the depth limit, it avoids getting permanently stuck down an infinitely long dead-end in the maze.
- **Time Complexity:** $O(b^d)$ where d is the depth of the shallowest goal.
- **Space Complexity:** $O(bd)$.
- **Advantages/Disadvantages:** It combines the memory efficiency of DFS with the completeness and optimality (if step costs are equal) of BFS. The disadvantage is that it re-generates top-level nodes multiple times, though in practice, this overhead is minimal compared to the memory saved.

Agent Design and Decision-Making

To solve a maze, a simple reflex agent will fail because it gets stuck in loops (it doesn't remember where it has been). The robot must be a **Model-Based Goal-Based Agent**.

- **Model-Based:** It must maintain an internal map of the paths it has already explored to avoid walking in circles.
- **Goal-Based:** It needs the explicit goal of "finding the exit" so it can apply search algorithms to plan a sequence of moves to reach that destination.

Most Effective Combination

The most effective combination is a **Model-Based Goal-Based Agent using Iterative Deepening Search (IDS)**.

Justification: Physical robots typically have limited onboard memory and processing power. While UCS guarantees the lowest-cost path, its exponential space complexity can quickly overwhelm the robot's memory in a complex maze. IDS provides a highly elegant solution: it guarantees the robot will find the shortest path (in terms of number of steps) while keeping memory requirements strictly linear ($O(bd)$). This ensures the robot can navigate even massive, complex mazes without crashing due to memory exhaustion.